The Besse volume of a contact form is bounded by the volume of the normalised form on the boundary

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Editorial note. The counted statement is the author's labelled lemma 'l:volume_Besse' (source lines 1227-1233); the proof environment begins on the next line after \end{lem} apart from one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's branched-covering equation, whose label is invoked inside the retained proof.

Editorial note. The retained preamble is a minimal editorial adapter supplying only the plain text abbreviations that the publisher class used to define, followed by the native author macro block of the pinned source.

Editorial note. Every label used inside the retained text is defined inside one of the retained spans, so the package declares no external reference.

Editorial scope. One original result of the pinned publisher article, counted once: the volume comparison for the Besse volume of a contact form (author label l:volume_Besse).

Allowed standard prerequisites (editorial).

- Besse volume. The Besse volume of a contact form is the measure induced by (lambda wedge dlambda)^n on the contact manifold.. It is the quantity bounded in the statement.
- The branched covering tilde-iota. The restriction of tilde-iota to the interior is an alpha-th fold branched covering map onto the solid cylinder, with the explicit model in the collar coordinates.. It is what allows the volume of the model to be compared with the volume form of the normalised contact form.
- The normalised form tilde-lambda_0. tilde-lambda_0 = dt + pr_1^* nu_0 is the normalised contact form on the interior, with Reeb field d/dt.. Its volume form is computed inside the retained proof.

$$\tilde{\iota}|_{\mathrm{int}(\Sigma)\times S^1} : \mathrm{int}(\Sigma) \times S^1 \to Y \setminus \gamma_1. \tag{1}$$

Lemma 0.1. *If (Y, λ_0) is a Besse contact 3-manifold whose closed Reeb orbits have minimal common period 1, its volume is equal to the negative of its Euler*

number. More precisely, if $\iota : \Sigma \to Y$ is a global surface of section as above, we have:

$$\mathrm{vol}(Y, \lambda_0) = \frac{1}{\alpha}\,\mathrm{area}(\Sigma, \omega_0) = -e(Y).$$

Proof. The contact volume form $\lambda_0 \wedge d\lambda_0$ is pulled back to

$$\tilde{\iota}^*(\lambda_0 \wedge \lambda_0) = \tilde{\lambda}_0 \wedge d\tilde{\lambda}_0 = dt \wedge \mathrm{pr}_1^* d\nu_0.$$

We recall that $\tilde{\iota}$ restricts to an α-th fold branched covering map (1). Therefore

$$\alpha\,\mathrm{vol}(Y, \lambda_0) = \alpha \int_Y \lambda_0 \wedge d\lambda_0 = \int_{\Sigma \times S^1} \tilde{\iota}^*(\lambda_0 \wedge d\lambda_0) = \int_{\Sigma \times S^1} dt \wedge \mathrm{pr}_1^* \omega_0 = \int_\Sigma \omega_0,$$

and the latter term is precisely $\mathrm{area}(\Sigma, \omega_0)$. Moreover, since $\iota|_{\partial\Sigma}$ is an orientation preserving $b\,p_0$-th fold covering map of the $1/\alpha_1$-periodic Reeb orbit γ_1, Stokes' theorem implies

$$\int_\Sigma \omega_0 = \int_{\partial\Sigma} \nu_0 = b\,p_0 \int_{\gamma_1} \lambda_0 = \frac{b\,p_0}{\alpha_1} = -\alpha\,e(Y). \qquad \square$$

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